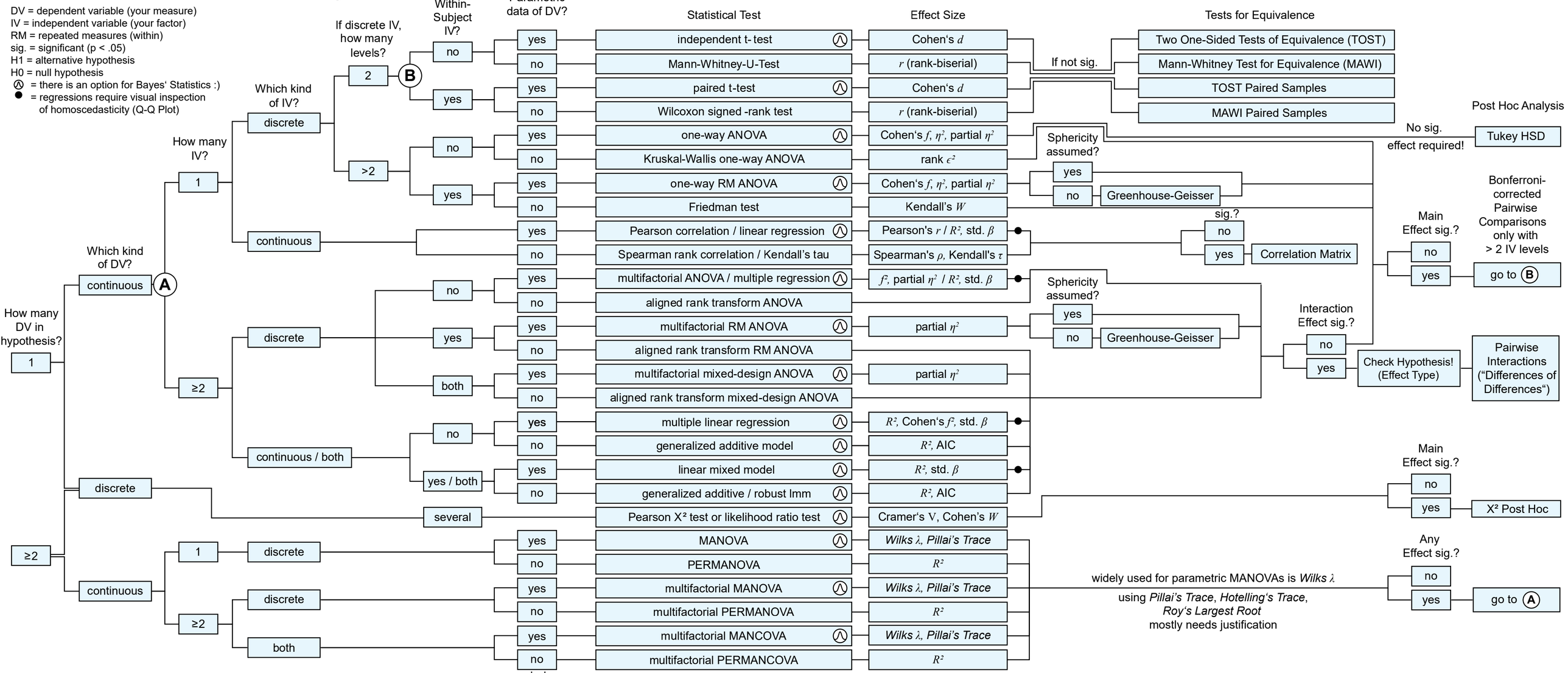
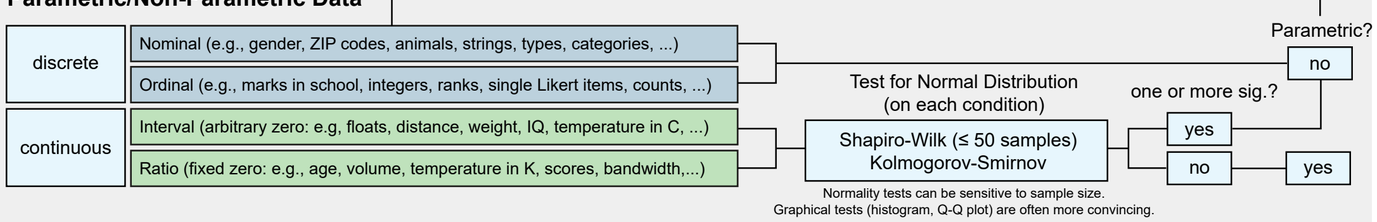
Inferential Statistics with Variance Analyses

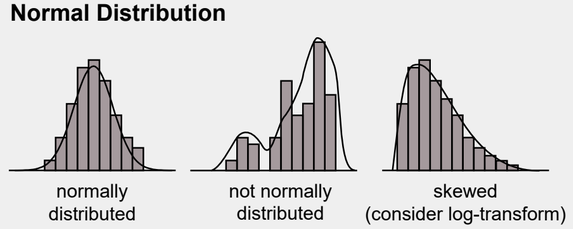
DV = dependent variable (your measure)
IV = independent variable (your factor)
RM = repeated measures (within)
sig. = significant ($p < .05$)
H1 = alternative hypothesis
H0 = null hypothesis
⚗ = there is an option for Bayes' Statistics :)
● = regressions require visual inspection of homoscedasticity (Q-Q Plot)



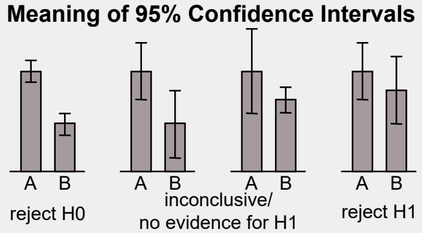
Parametric/Non-Parametric Data



Normal Distribution



Meaning of 95% Confidence Intervals



Interpretation of Effect Sizes

	negligible	small	medium	large
Cohen's d	0.20	0.50	0.80	
r, Cohen's/Kendall's W	0.10	0.30	0.50	
Cohen's f	0.10	0.25	0.40	
Cohen's f^2	0.02	0.15	0.35	
η^2	0.02	0.13	0.26	
rank ϵ^2, partial η^2	0.01	0.06	0.14	

Kendall (1939), Cohen (1988)

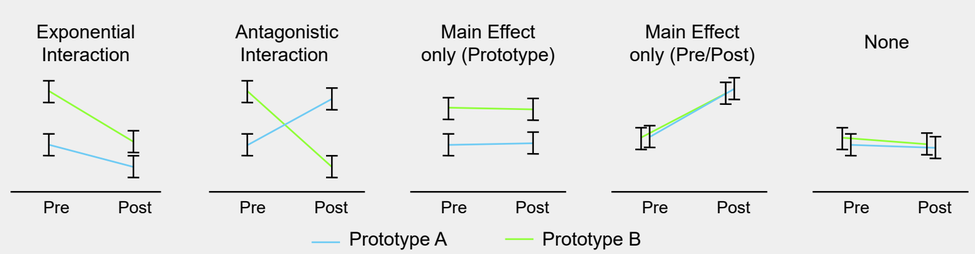
Some effect sizes can be negative.
This does not affect the interpretation.

Interpretation of Correlations / Fits

	negligible	weak	moderate	strong	perfect
Pearson's r, ρ, τ, R^2, std. β	.20	.50	.80	=1	
Cramer's V (df=1)	.10	.30	.50	=1	
Cramer's V (df=2)	.07	.21	.35	=1	
Cramer's V (df=3)	.06	.17	.29	=1	

Kendall (1939), Cohen (1988), Gravetter & Wallnau (2004)

Effect Types



Valentin Schwind
Released under GNU GPL 3.0
https://github.com/valentin-schwind/statistics-decision-tree